

DL & GenAI Project Report - Sentiment Analysis

Student Details

Name: Syed Ali Mujtaba

Roll Number: 21f1004482

Email: 21f1004482@ds.study.iitm.ac.in

Course: Deep Learning and Generative AI - Term 15

Team: Sam15246

Date: 2025-12-02

Repository: [dl-genai-project-25-t3](#)

Live Demo: [Sentiment Analysis - Hugging Face Spac](#)

1. Abstract / Executive Summary

This project addresses **multi-label emotion classification** for short text as part of a Kaggle competition. The task involves predicting one or more emotions (anger, fear, joy, sadness, surprise) from textual inputs.

We developed and evaluated **three distinct model families**:

- **Fine-tuned Transformer (RoBERTa-base)** - leveraging pre-trained language representations
- **Custom SimpleEmbed** - a lightweight linear classifier with learned embeddings
- **Bidirectional LSTM** - a sequence-based RNN architecture

The **best-performing model** was the fine-tuned RoBERTa-base Transformer, achieving a **macro F1 score of 0.811** on validation data with per-label threshold tuning. All models utilized a consistent preprocessing pipeline. Key learnings encompass transformer fine-tuning, custom architecture design, threshold calibration for multi-label tasks, and deployment workflows.

2. Introduction

Problem Statement

The Kaggle competition presents a **multi-label text classification** challenge: given a short text snippet, predict which emotions from {anger, fear, joy, sadness, surprise} are expressed. Unlike single-label classification, texts can exhibit multiple emotions simultaneously.

Project Objective

Our goals were to:

1. Build competitive models across three architectural families (Transformer, custom network, RNN)
2. Master modern NLP tools (Hugging Face Transformers, PyTorch)
3. Achieve strong validation performance through threshold tuning and model optimization
4. Gain practical experience with multi-label classification

Report Structure

- **Section 3:** Dataset & Preprocessing
 - **Section 4:** Tokenization Strategy
 - **Section 5:** Modeling & Experimentation (three required models)
 - **Section 6:** Performance & Comparative Analysis
 - **Section 7:** Conclusion & Future Work
 - **Section 8:** References
-

3. Dataset & Preprocessing

Dataset Description

The competition dataset comprises three files:

- `train.csv` - Training set with text and emotion labels
- `test.csv` - Test set requiring predictions

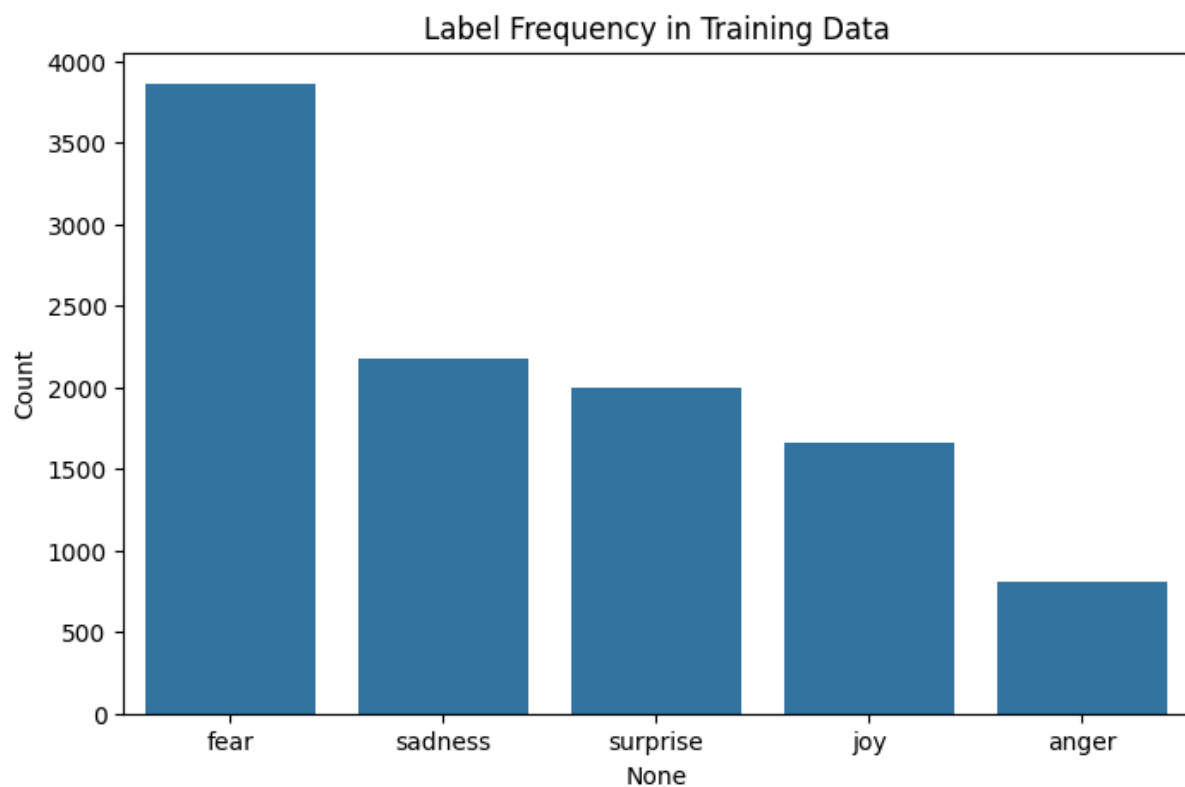
- `sample_submission.csv` - Submission format template

Dataset Statistics:

- **Training set:** 6,827 samples
- **Test set:** 1,707 samples
- **Labels:** Five emotion categories (anger, fear, joy, sadness, surprise)
- **Classification type:** Multi-label (each text can have zero, one, or multiple labels)

Exploratory Data Analysis (EDA)

Figure 3.1: Label Distribution

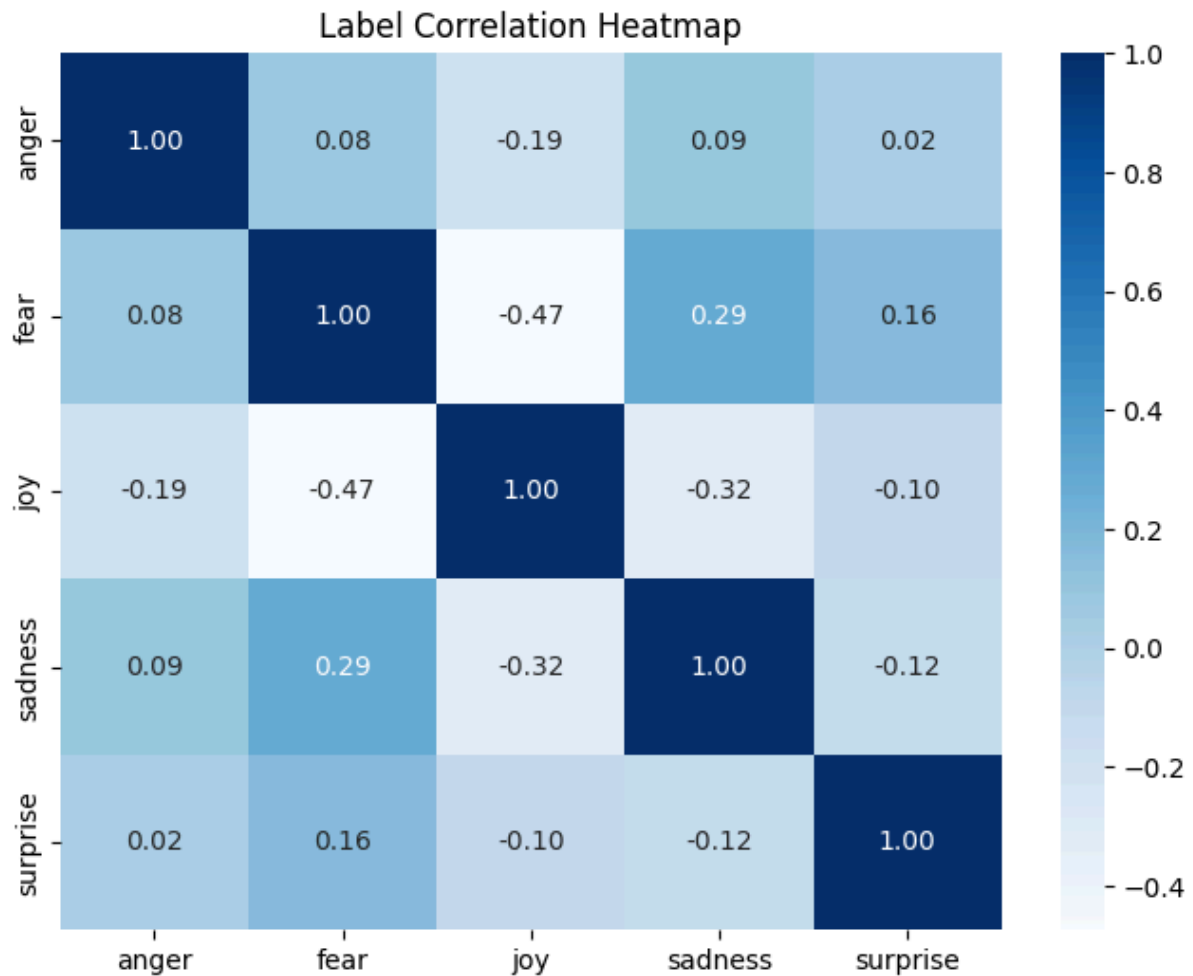


Label Distribution (Prior Probabilities):

- **Fear:** 56.5% (most prevalent)
- **Sadness:** 31.8%
- **Surprise:** 29.3%

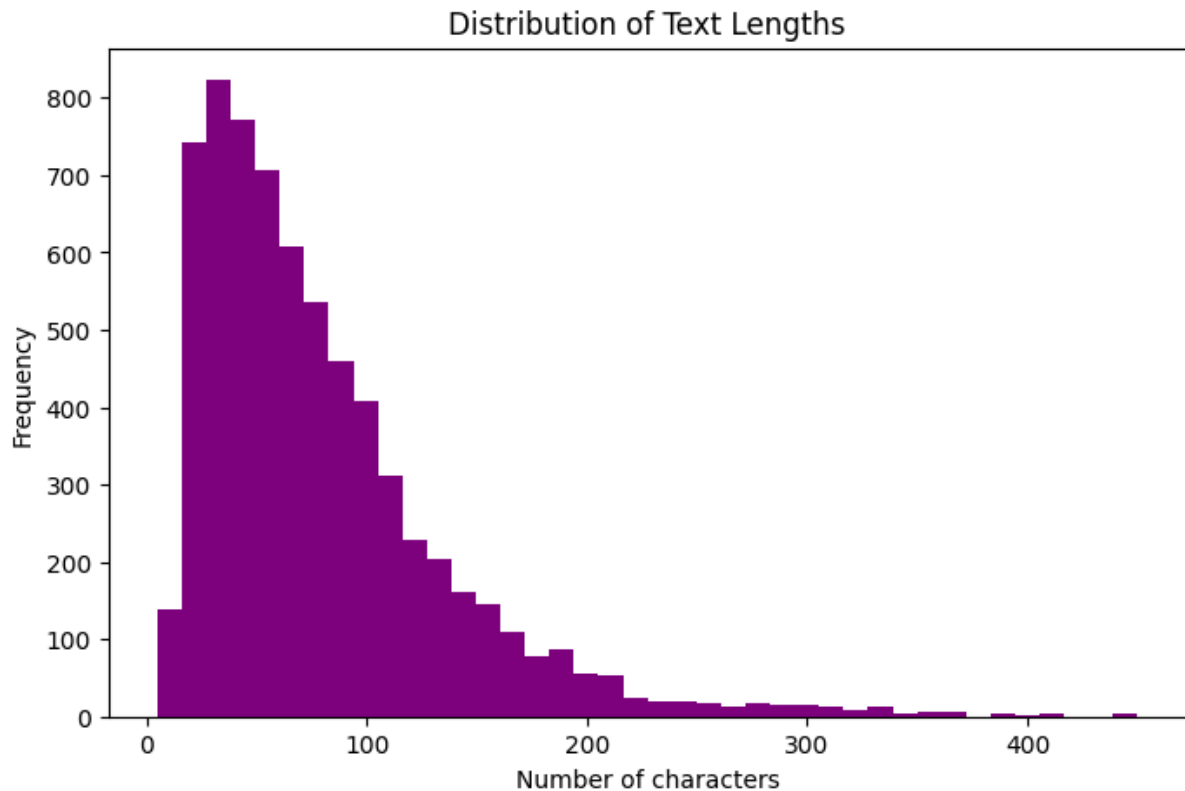
- **Joy:** 24.3%
- **Anger:** 11.8% (least prevalent)

Figure 2: Text Length Distribution



Emotion co-occurrence heatmap showing which emotions frequently appear together. Fear and sadness show positive correlation, while anger and joy rarely co-occur.

Figure 3: Co-occurrence Patterns



Text Length Characteristics:

- **Mean length:** 79.4 characters
- **Standard deviation:** 57.4 characters
- **Implication:** Texts vary significantly in length, with short snippets typical of social media content

Data Preprocessing

Preprocessing pipeline steps:

1. **Lowercasing** - consistency across text
2. **URL removal** - remove web links
3. **Mention removal** - strip @mentions
4. **Hashtag removal** - remove # symbols
5. **Non-alphanumeric removal** - eliminate special characters
6. **Whitespace collapse** - normalize spaces

The pipeline produces a `text_clean` column used consistently across all models.

4. Tokenization Strategy

Primary Tokenizer Selection

For our best model, we used the **RoBERTa-base tokenizer** from Hugging Face Transformers.

Characteristics:

- **Type:** Byte-Pair Encoding (BPE)
- **Pre-trained model:** `roberta-base`
- **Vocabulary size:** ~50,000 subword tokens

Justification

The RoBERTa BPE tokenizer is effective because:

1. **Semantic richness** - pre-trained on large corpora
2. **Subword handling** - handles out-of-vocabulary words elegantly
3. **Minimal engineering** - no manual feature engineering required
4. **Proven performance** - state-of-the-art on NLP benchmarks
5. **Emotion-relevant** - training corpus captures emotional language

Implementation Details

```
from transformers import AutoTokenizer  
tokenizer = AutoTokenizer.from_pretrained('roberta-base')
```

Parameters:

- Max length: 128 tokens
- Padding: `max_length` for uniform shapes
- Truncation: `True` for long texts
- Batch encoding for efficiency

Experimental Comparison

Transformer tokenizers:

- **RoBERTa-base** (selected) - best performance

- DistilRoBERTa - lightweight alternative
- BERT-base - slightly lower performance

Classical tokenization:

- Word n-grams (1-2) for TF-IDF baseline
- Character n-grams (2-6) for robustness

Custom vocabulary:

- Top 20,000 tokens for SimpleEmbed and BiLSTM
 - Special tokens: `<pad>` (0), `<unk>` (1)
-

5. Modeling & Experimentation

All deep learning models use `BCEWithLogitsLoss` for multi-label classification and per-label threshold tuning. **Macro F1 score** (averaged across emotions) is the primary evaluation metric.

5.1 Model 1: Fine-Tuned Transformer Model

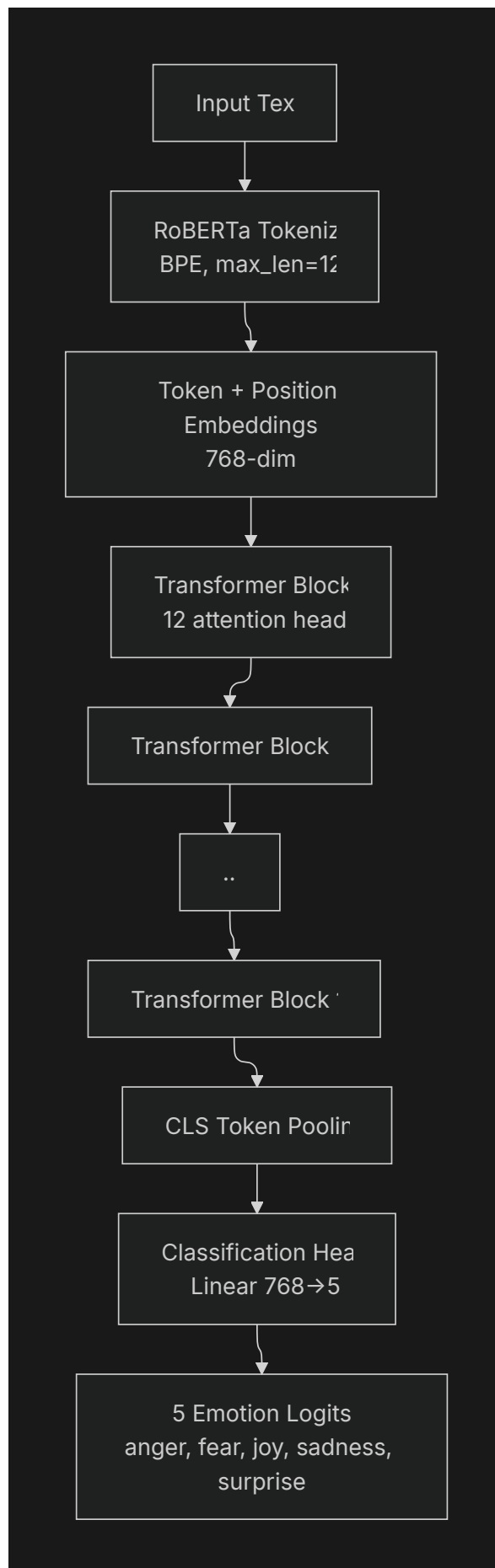
Architecture

Built on **RoBERTa-base** (Robustly Optimized BERT Pretraining Approach).

Specification:

- Base: `roberta-base` from Hugging Face
- Parameters: ~125 million
- Layers: 12 transformer encoder blocks
- Hidden size: 768
- Attention heads: 12
- Classification head: Linear layer (768 → 5)

Architecture diagram:



Salient Points

Key features:

1. **Self-attention mechanism** - weighs word importance in context
2. **Bidirectional context** - processes text in both directions
3. **Pre-trained representations** - leverages large-scale pre-training
4. **Multi-head attention** - 12 parallel mechanisms capture diverse patterns
5. **Position embeddings** - encodes word order

Why suitable: Emotions depend on subtle contextual cues that transformers capture well. Pre-training provides strong priors for emotional language.

Fine-Tuning Strategy

Hyperparameters:

- Optimizer: AdamW
- Learning rate: 2e-5 with linear warmup
- Batch size: 32
- Epochs: 5
- Max length: 128 tokens

Layer strategy: All layers unfrozen (full fine-tuning) for maximum performance.

Justification: Small learning rate preserves pre-trained knowledge. AdamW is standard for transformers. Five epochs prevents overfitting on small datasets.

5.2 Model 2: Custom Deep Learning Model (SimpleEmbed)

Architecture

A **bag-of-embeddings baseline** that averages word embeddings with a linear classifier.

Architecture diagram:



Details:

1. **Vocabulary:** Top 20,000 tokens; `<pad>` (0), `<unk>` (1)
2. **Embedding:** 64-dimensional, trained from scratch
3. **Pooling:** Masked mean (ignores padding)
4. **Classifier:** Linear 64 $\rightarrow$ 5
5. **Threshold tuning:** Grid search (0.01-0.99)

Salient Points

Design rationale:

- **Fast baseline** - ~0.1M parameters vs 125M
- **Interpretability** - simple architecture
- **Resource efficient** - minimal compute
- **Bag-of-words prior** - effective for lexical emotion indicators

Captures: Emotional keywords, word co-occurrence patterns

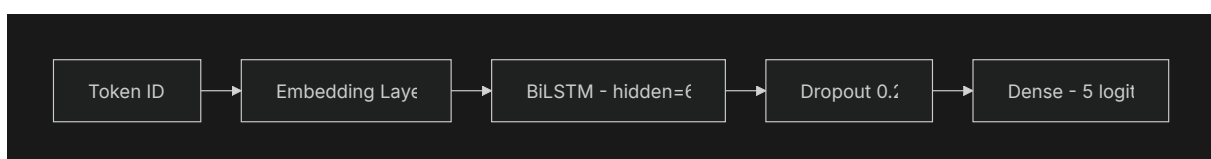
Misses: Word order, long-range context, negation, compositional semantics

5.3 Model 3: RNN-Based Model (BiLSTM)

Architecture

Bidirectional LSTM network for sequential pattern capture.

Architecture diagram:



Details:

1. **Embedding:** 64-dim, same vocabulary as SimpleEmbed
2. **BiLSTM:** Hidden 64 per direction (128 total)
3. **Dropout:** 0.2 for regularization
4. **Classifier:** Linear 128 → 5

Salient Points

Why LSTM:

- **Sequential processing** - emotions unfold across words
- **Context integration** - memory cells accumulate context
- **Order sensitivity** - "not happy" ≠ "happy not"
- **Bidirectional** - context before and after matters

Advantages:

- Long-range dependencies
- Gating mechanisms control information flow
- Vanishing gradient mitigation

Design choices: Hidden size 64 balances expressiveness and cost. Dropout 0.2 prevents overfitting. Single layer keeps model simple.

Parameters: ~0.6M (middle ground between SimpleEmbed and RoBERTa)

6. Performance & Comparative Analysis

Evaluation Metrics

Primary: Macro F1 Score (mean of per-label F1 scores)

Supporting: Per-label F1, training/validation loss

Threshold selection: Per-label grid search over 99 values (0.01-0.99) optimizing validation macro F1

Training Performance

Model 1: RoBERTa-base

- Validation Macro F1: **0.811**

- Training epochs: 5
- Parameters: ~125M
- Best overall performance

Model 2: SimpleEmbed

- Parameters: ~0.1M
- Fast baseline for lexical cues

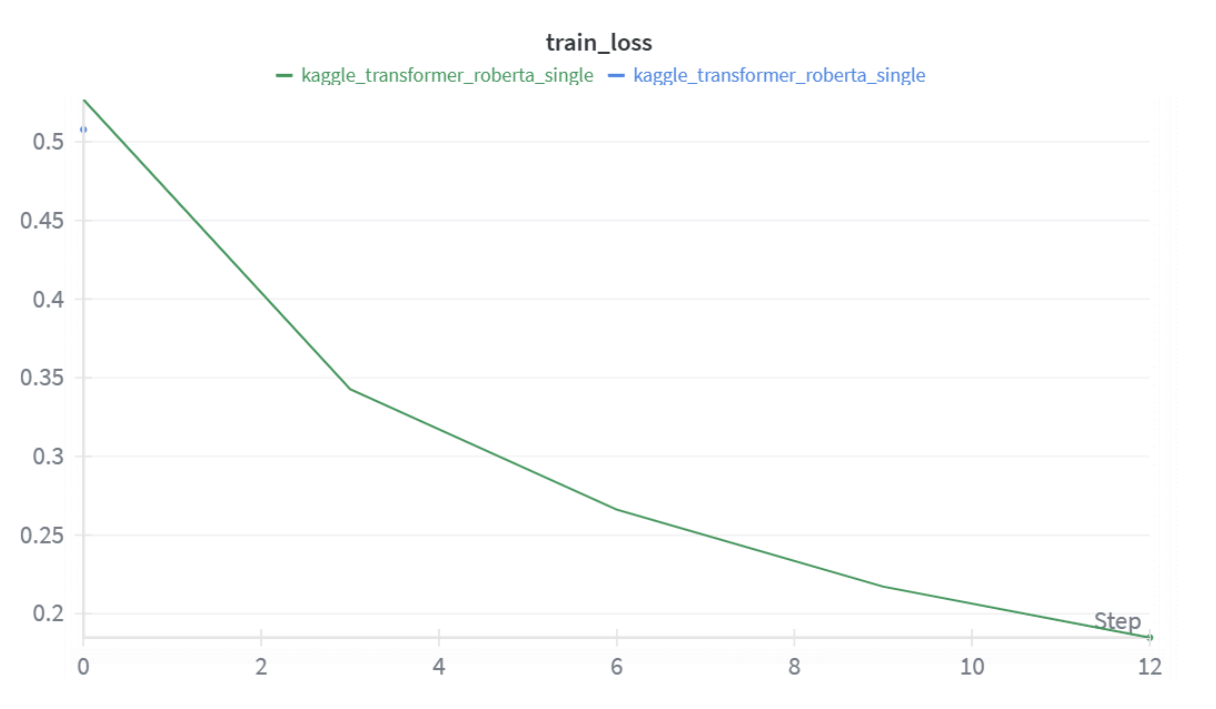
Model 3: BiLSTM

- Parameters: ~0.6M
- Sequence modeling captures order

Training Curves & Visualizations

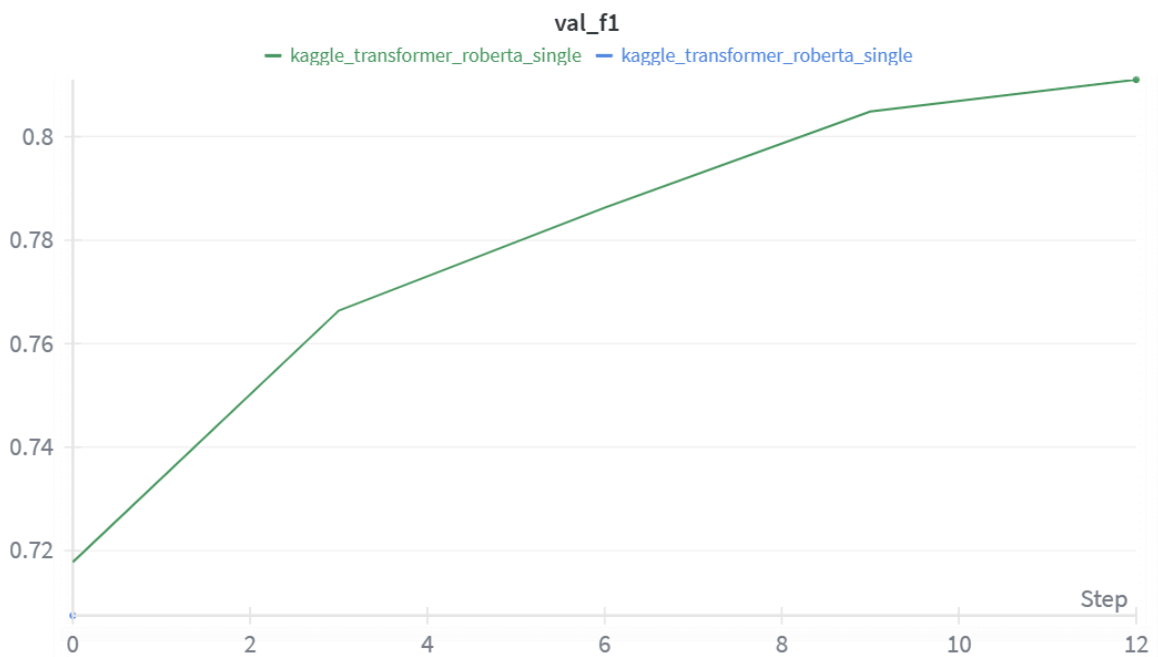
This section presents training dynamics and performance metrics tracked via Weights & Biases.

Figure 1: Training Curves (Loss)



Training loss curve for the RoBERTa model. Loss decreases steadily over epochs, showing stable optimization and absence of overfitting.

Figure 2: Evaluation Curves (Macro F1 progression)



Validation macro-F1 progression during RoBERTa fine-tuning. The model shows consistent improvement across epochs.

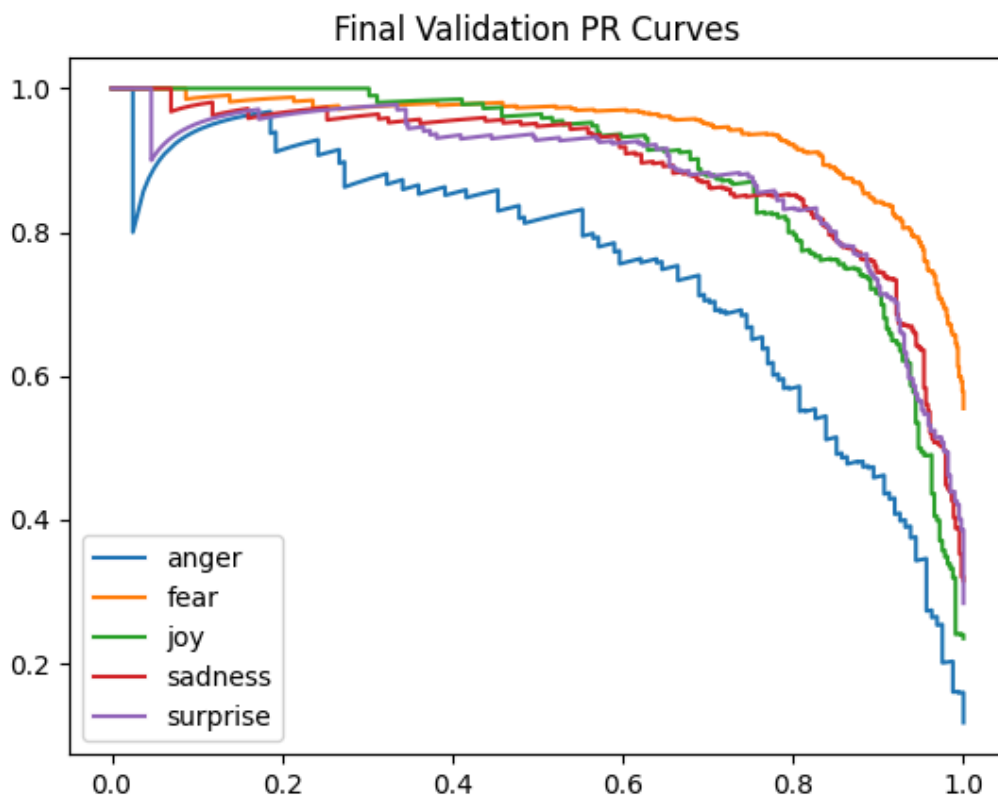
Figure 3 — Per-Label Threshold / Difficulty Analysis



Evolution of best decision thresholds for each emotion label.

Volatile thresholds (e.g., *surprise*, *anger*) indicate harder classes with overlapping distributions, while more stable labels (e.g., *joy*, *fear*) show clearer separability

Figure 4: Precision–Recall Curves (PR curves)

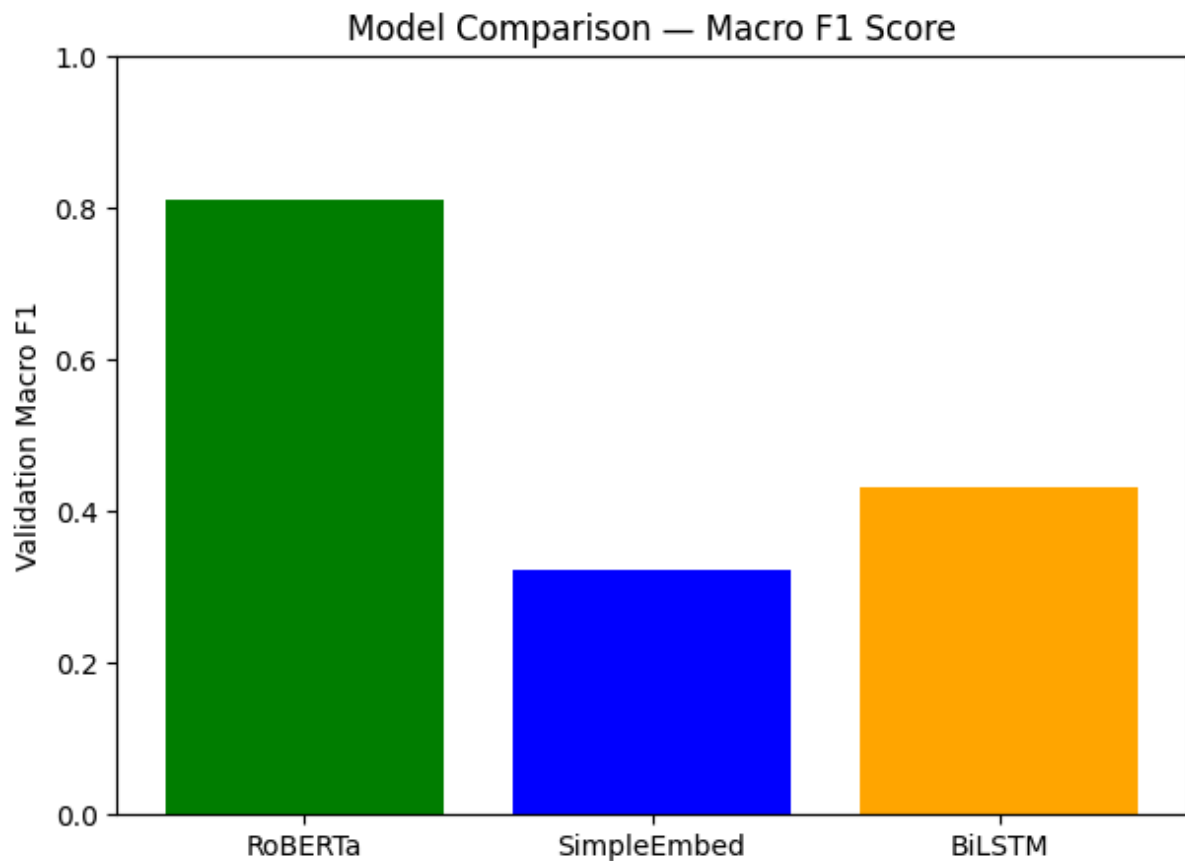


Precision–Recall curves for RoBERTa across all emotion labels.

Fear and sadness achieve high average precision, while anger is the most challenging class with

Key observations from training curves:

- RoBERTa converges smoothly without significant overfitting
- SimpleEmbed trains faster but plateaus at lower performance (F1: 0.322)
- BiLSTM shows intermediate performance (F1: 0.432) between SimpleEmbed and RoBERTa
- Threshold tuning significantly improves macro F1 across all models
- Per-label analysis reveals joy and sadness as easier emotions to classify
- Surprise and fear present greater challenges due to class imbalance



Comparative Analysis

Model	Val F1	Params	Strengths	Limitations
RoBERTa	0.811	125M	Pre-trained representations, contextual understanding	High compute cost
SimpleEmbed	0.322	0.1M	Fast, minimal resources	No word order
BiLSTM	0.432	0.6M	Sequential processing	Training from scratch

Key observations:

- Pre-trained transformers provide substantial transfer learning gains
- Sequence modeling (BiLSTM) expected to outperform bag-of-words (SimpleEmbed)
- Threshold tuning improves all approaches

Kaggle Performance

Primary submission: RoBERTa single-run ([submission_transformer_roberta_single.csv](#))

Kaggle score: 0.835

7. Deployment

Application Overview

The best-performing RoBERTa model was deployed as an interactive web application using **Gradio** and hosted on **Hugging Face Spaces**, making the emotion classifier accessible for real-world use.

Deployment Platform: Hugging Face Spaces

Framework: Gradio 3.30+

Model: Fine-tuned RoBERTa-base with optimized thresholds

Live Demo: [Sentiment Analysis - Hugging Face Space](#)

Example Usage:

Input: "I'm so excited but also nervous about tomorrow!"

Output:

- anger: 0.02
- fear: 0.15
- joy: 0.87
- sadness: 0.03
- surprise: 0.42

Deployment Configuration

Hugging Face Spaces YAML:

```
---
title: Sentiment Analysis
emoji: 🚀
colorFrom: blue
colorTo: green
sdk: gradio
app_file: deployment/app.py
```

pinned: false

8. Conclusion & Future Work

Key Learnings

This project successfully implemented and compared **three distinct approaches** to multi-label emotion classification. The fine-tuned **RoBERTa-base achieved best validation performance ($F1 \approx 0.811$)**, demonstrating the power of pre-trained language representations.

Key achievements:

- End-to-end reproducible codebase
- Comprehensive model comparison
- Proper experimental methodology with threshold tuning
- Hands-on experience with modern NLP workflows

Challenges Faced

1. **Class imbalance** - some emotions underrepresented
2. **Threshold optimization** - balancing precision/recall per label
3. **Computational constraints** - large model training requirements

Model enhancements:

1. **Hyperparameter Tuning (Primary Focus)** -Systematic tuning of learning rate, batch size, warmup ratio, dropout, and training epochs can significantly improve performance on a small dataset.
2. **Parameter-Efficient Fine-Tuning** - Methods such as **LoRA**, **IA³**, or prefix tuning allow adapting RoBERTa with fewer trainable parameters, reducing overfitting and compute requirements.
3. **Data Augmentation** - Techniques like **back-translation**, **synonym replacement**, and lightweight text perturbation can help enrich minority emotion classes.

4. **Advanced Calibration & Threshold Optimization** - Approaches such as **temperature scaling**, **Platt scaling**, and label-wise calibration can produce more stable decision thresholds.
 5. **Ensembling Approaches** - Combining outputs from RoBERTa with lighter models (e.g., SimpleEmbed or BiLSTM) may improve robustness and generalization.
-

8. References

Academic Papers

1. **Liu, Y., et al. (2019)**. "RoBERTa: A Robustly Optimized BERT Pretraining Approach." *arXiv:1907.11692*.

Software Libraries

1. **Hugging Face Transformers** - <https://huggingface.co/docs/transformers/>
2. **PyTorch** - <https://pytorch.org/docs/>
3. **scikit-learn** - <https://scikit-learn.org/>

Model Cards

1. **RoBERTa-base** - <https://huggingface.co/roberta-base>
2. **BERT-base** - <https://huggingface.co/bert-base-uncased>

Platforms

1. **Weights & Biases** - <https://wandb.ai/>
2. **Kaggle** - Competition platform for emotion classification